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Cli(command line interface):

A **Command Line Interface (CLI)** is a type of user interface where the user interacts with a computer by typing commands into a text-based terminal or command prompt. The system executes tasks based on these commands instead of using graphical elements.

Code used to understand : Python(cli delete interface)

Code:

```
import os

import sys


def delete_file(filename):

  try:

    os.remove(filename)

    print(f"File '{filename}' deleted successfully.")

  except FileNotFoundError:

    print(f"Error: File '{filename}' not found.")

  except Exception as e:

    print(f"An error occurred: {e}")


if __name__== "__main__":

  if len(sys.argv) != 2:

    print("Usage: python delete_file_cli.py <filename>")

  else:

    delete_file(sys.argv[1])
```

This CLI program allows the user to delete a file by providing the filename as a command-line argument. It uses system-level operations to remove the file and handles errors such as missing files or incorrect input.

Output terminal:

```
Microsoft Windows [Version 10.0.26200.7848]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shivam>cd OneDrive\Desktop\daa

C:\Users\shivam\OneDrive\Desktop\daa>py delete_file_cli.py
Usage: python delete_file_cli.py <filename>

C:\Users\shivam\OneDrive\Desktop\daa>echo hello > test.txt

C:\Users\shivam\OneDrive\Desktop\daa>py delete_file_cli.py test.txt
File 'test.txt' deleted successfully.

C:\Users\shivam\OneDrive\Desktop\daa>_
```

Thereby the terminal which created “hello.test “ was deleted successfully using cli command line.

What is an gui?

A **Graphical User Interface (GUI)** is a user interface that allows users to interact with a system using visual elements such as windows, icons, buttons, and menus. It is more user-friendly and does not require memorizing commands.

Code used : Python (gui delete user interface).

Code implementation:

```
import os

import tkinter as tk

from tkinter import filedialog, messagebox


def delete_file():

  filename = filedialog.askopenfilename(title="Select a file to delete")

  if filename:

    try:

      os.remove(filename)

      messagebox.showinfo("Success", f"File '{filename}' deleted successfully.")

    except FileNotFoundError:
```

```
        messagebox.showerror("Error", f"File '{filename}' not found.")
    except Exception as e:
        messagebox.showerror("Error", f"An error occurred: {e}")



root = tk.Tk()
root.title("File Deleter")



delete_button = tk.Button(root, text="Delete a File", command=delete_file)
delete_button.pack(pady=20)



root.mainloop()
```

Output terminal:

```
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shivam>cd OneDrive\Desktop\daa

C:\Users\shivam\OneDrive\Desktop\daa>py delete_file_gui.py
```

what it does?

This program creates a **Graphical User Interface (GUI)** using Python where the user can **select and delete a file by clicking a button**, instead of typing commands.

A button appears when command is runned then the file name is entered its directed to folder which was saved in notepad and deletes it successfully thereby in command terminal message displays successfully as “deleted file gui.py”

Voice recognition user interface(vui):

What is an vui?

A **Voice User Interface (VUI)** is a type of interface where users interact with a system using voice commands. The system recognizes spoken input and responds accordingly, enabling hands-free operation.

Code used:Python (displaying message using vui)

Code implementation:

```
import speech_recognition as sr

import pyttsx3

import os

import datetime

import webbrowser


engine = pyttsx3.init()

engine.setProperty('rate', 170)


def speak(text):

  print("Assistant:", text)

  engine.say(text)

  engine.runAndWait()


def listen():

  r = sr.Recognizer()

  with sr.Microphone() as source:

    print("\nListening...")

    r.adjust_for_ambient_noise(source, duration=1)
```

```
        audio = r.listen(source)

    try:
        command = r.recognize_google(audio)
        print("You said:", command)
        return command.lower()
    except sr.UnknownValueError:
        speak("Sorry, I did not understand.")
        return ""
    except sr.RequestError:
        speak("Speech service is not available.")
        return ""

def execute_command(command):
    if "hello" in command:
        speak("Hello! How can I help you?")

    elif "open notepad" in command:
        speak("Opening Notepad")
        os.system("notepad")

    elif "open calculator" in command:
        speak("Opening Calculator")
        os.system("calc")

    elif "open chrome" in command:
        speak("Opening Chrome")
        os.system("start chrome")
```

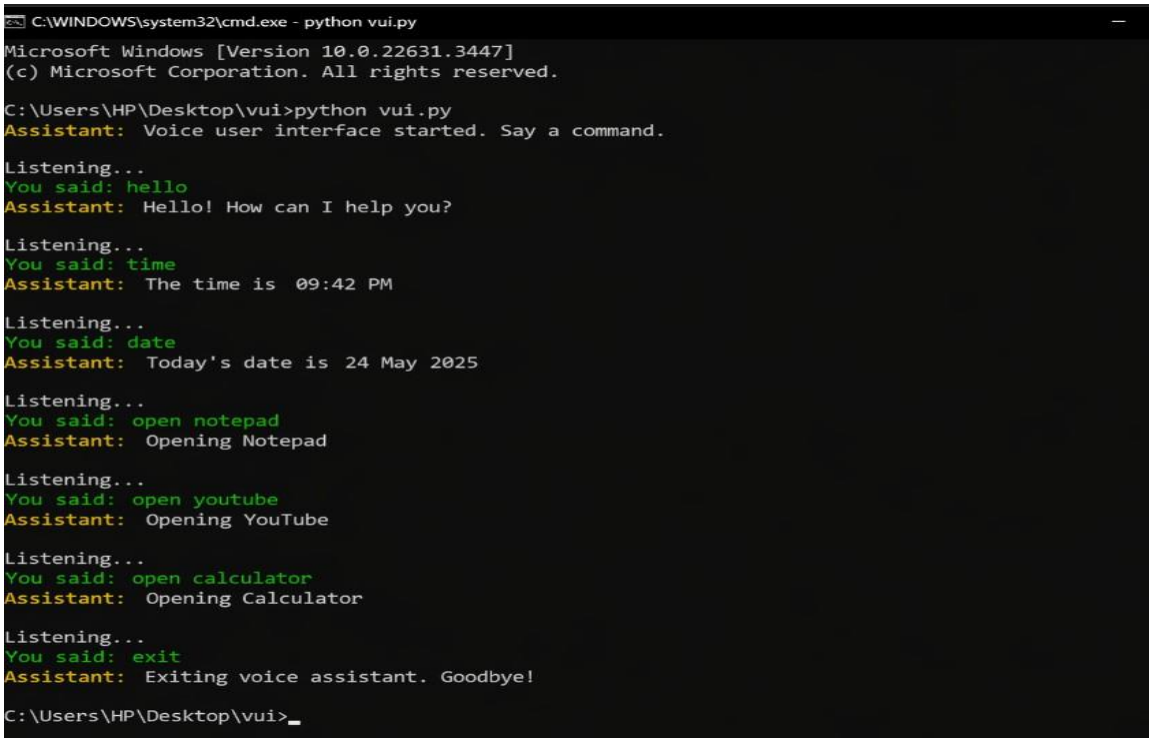
```
    elif "open youtube" in command:
        speak("Opening YouTube")
        webbrowser.open("https://www.youtube.com")

    elif "time" in command:
        current_time = datetime.datetime.now().strftime("%I:%M %p")
        speak(f"The time is {current_time}")

    elif "date" in command:
        today = datetime.datetime.now().strftime("%d %B %Y")
        speak(f"Today's date is {today}")

    elif "exit" in command or "stop" in command:
        speak("Exiting voice assistant. Goodbye!")
        return False

    elif command != "":
        speak("Command not recognized.")

    return True

def main():
    speak("Voice user interface started. Say a command.")
    running = True
    while running:
        command = listen()
        running = execute_command(command)
```

```
if __name__== "__main__":
  main
```

what it does?

This program creates a **Voice User Interface (VUI)** where the user gives **voice commands through a microphone**, and the system **recognizes, processes, and responds using speech and actions**.

Output terminal:



```
C:\WINDOWS\system32\cmd.exe - python vui.py
Microsoft Windows [Version 10.0.22631.3447]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP\Desktop\vui>python vui.py
Assistant: Voice user interface started. Say a command.

Listening...
You said: hello
Assistant: Hello! How can I help you?

Listening...
You said: time
Assistant: The time is 09:42 PM

Listening...
You said: date
Assistant: Today's date is 24 May 2025

Listening...
You said: open notepad
Assistant: Opening Notepad

Listening...
You said: open youtube
Assistant: Opening YouTube

Listening...
You said: open calculator
Assistant: Opening Calculator

Listening...
You said: exit
Assistant: Exiting voice assistant. Goodbye!

C:\Users\HP\Desktop\vui>
```

<u>Conclusion:</u>

Therefore the application of vui,cli,gui was learned successfully implemented using python and commands were understood and application was successfully displayed in the terminal and was done successfully with displaying messages.